8h. Type-Stability Gotchas

Martin Alfaro
PhD in Economics

INTRODUCTION

This section presents subtle scenarios where type instabilities arise. Since the root cause of the type instability isn't immediately obvious, we refer to these cases as "gotchas" and offer guidance on how to address them. To ensure self-containment, we revisit some examples previously discussed, but now providing recommendations for their mitigation.

GOTCHA 1: INTEGERS AND FLOATS

`Int64` and `Float64` are distinct types. Even though Julia promotes integers to floating-point numbers in many contexts, mixing them can still inadvertently introduce type instability.

To illustrate this, consider a function `foo` that takes a numeric variable `x` as its argument and performs two tasks: it first defines a variable `y` that replaces `x`'s negative values with zero, and then it executes an operation based on the resulting `y`.

In the following, we implement `foo` with an approach that suffers from type instability, and another one that addresses the issue.

```julia
function foo(x)
    y = (x < 0) ?  0  :  x

    return [y * i for i in 1:100]
end

@code_warntype foo(1)      # type stable
@code_warntype foo(1.0)    # type UNSTABLE
```

```julia
function foo(x)
    y = (x < 0) ?  zero(x)  :  x

    return [y * i for i in 1:100]
end

@code_warntype foo(1)      # type stable
@code_warntype foo(1.0)    # type stable
```

The first implementation uses the literal `0`, whose type is `Int64`. If `x` is also `Int64`, no type instability arises. However, if `x` is `Float64`, the compiler treats `y` as potentially `Int64` or `Float64`, thus causing type instability. [1]

Note, though, that Julia can generally handle combinations of `Int64` and `Float64` quite effectively. Thus, this type instability wouldn't be a significant problem if the operation calls `y` only once. Indeed, `@code_warntype` in this case would simply issue a yellow warning, hinting at potential for optimization but not a severe performance bottleneck. However, `foo` in our example repeatedly performs an operation involving `y`, incurring the cost of type instability multiple times. As a result, `@code_warntype` issues a red warning, indicating a more serious performance issue.

The second tab proposes a **solution** for this scenario. It introduces a function that returns the zero element of `x`'s type, instead of `0`. In this way, `y` is created ensuring that types won't be mixed.

This approach to solving type instability can be extended to values different from zero, by use of the function `convert(typeof(x), <value>)` or `oftype(x, <value>)`. Both convert `<value>` to the same type as `x`. For instance, below we reimplement `foo` using the value `5` instead of `0`.

```
function foo(x)
    y = (x < 0) ?  5  :  x

    return [y * i for i in 1:100]
end

@code_warntype foo(1)       # type stable
@code_warntype foo(1.0)     # type UNSTABLE
```

```
function foo(x)
    y = (x < 0) ?  convert(typeof(x), 5)  :  x

    return [y * i for i in 1:100]
end

@code_warntype foo(1)       # type stable
@code_warntype foo(1.0)     # type stable
```

```
function foo(x)
    y = (x < 0) ?  oftype(x, 5)  :  x

    return [y * i for i in 1:100]
end

@code_warntype foo(1)       # type stable
@code_warntype foo(1.0)     # type stable
```

GOTCHA 2: COLLECTIONS OF COLLECTIONS

In data analysis, it's common to manipulate *collections of collections*, where one data structure nests others inside it. A well-known example in Julia is the `DataFrames` package, which organizes data into columns that represent multiple variables. In this case, each column is a collection itself, with the full set of columns acting as another collection. Since we haven't introduced this package, we'll consider a simpler analogous structure: a vector of vectors, whose type is `Vector{Vector}`.

The primary benefit of using `Vector{Vector}` is its inherent flexibility. Since it imposes no constraints on the element types stored in the inner vectors, each vector can hold whatever data you need: strings, integers, floating-point numbers, or even mixture of these. This makes it easy to represent heterogeneous datasets, without committing to a rigid schema.

That same flexibility, however, comes with a cost. The type system only knows that each element is some vector, with no information about the concrete element type inside each inner vector. Without that information, the compiler can't infer types when your code operates on these vectors, thus leading to type instability.

To see this more concretely, imagine a vector `data` whose elements are themselves vectors. Suppose we write a function `foo` that receives `data` and performs some computation on one of its inner vectors, `vec2`. As shown in the first tab, the compiler can determine that `vec2` is a vector, but it can't deduce the type of its elements. Consequently, calls to `foo` become type-unstable.

A simple and effective way to **address** this issue appears in the second tab. The solution is to introduce a barrier function that takes the inner vector `vec2` as its argument. The barrier function then rectifies the type instability by attempting to identify a concrete type for `vec2`.

```julia
vec1 = ["a", "b", "c"] ; vec2 = [1, 2, 3]
data = [vec1, vec2]

function foo(data)
    for i in eachindex(data[2])
        data[2][i] = 2 * i
    end
end

@code_warntype foo(data)            # type UNSTABLE
```

```julia
vec1 = ["a", "b", "c"] ; vec2 = [1, 2, 3]
data = [vec1, vec2]

foo(data) = operation!(data[2])

function operation!(x)
    for i in eachindex(x)
        x[i] = 2 * i
    end
end

@code_warntype foo(data)            # barrier-function solution
```

Note that the second tab defines the barrier function in-place. This means that the function directly updates the contents of the inner vector `vec2`, rather than creating a new copy. Because `vec2` is part of the larger structure data, the outer structure `data` is updated as well. This approach is typical in data-analysis workflows, where the objective is to transform an existing dataset, rather than allocate a fresh one every time a change is applied.

GOTCHA 3: BARRIER FUNCTIONS

Barrier functions are an effective technique to mitigate type instabilities. However, keep in mind that **the parent function may remain type unstable**. When this occurs and instability isn't resolved before executing a repeated operation, the associated performance penalty will be incurred multiple times.

To illustrate this point, let's revisit the last example involving a vector of vectors. Below, we present two incorrect approaches to using a barrier function, followed by a demonstration of its proper application.

```
vec1 = ["a", "b", "c"] ; vec2 = [1, 2, 3]
data = [vec1, vec2]

operation(i) = (2 * i)

function foo(data)
    for i in eachindex(data[2])
        data[2][i] = operation(i)
    end
end

@code_warntype foo(data)            # type UNSTABLE
```

```
vec1 = ["a", "b", "c"] ; vec2 = [1, 2, 3]
data = [vec1, vec2]

operation!(x,i) = (x[i] = 2 * i)

function foo(data)
    for i in eachindex(data[2])
        operation!(data[2], i)
    end
end

@code_warntype foo(data)            # type UNSTABLE
```

```julia
vec1 = ["a", "b", "c"] ; vec2 = [1, 2, 3]
data = [vec1, vec2]

function operation!(x)
    for i in eachindex(x)
        x[i] = 2 * i
    end
end

foo(data) = operation!(data[2])

@code_warntype foo(data)            # barrier-function solution
```

GOTCHA 4: INFERENCE IS BY TYPE, NOT BY VALUE

Julia's compiler generates method instances solely on the basis of types, without taking actual values into account. To demonstrate this, consider the following example.

```julia
function foo(condition)
    y = condition ?  2.5  :  1

    return [y * i for i in 1:100]
end

@code_warntype foo(true)          # type UNSTABLE
@code_warntype foo(false)         # type UNSTABLE
```

At first glance, we might erroneously conclude that `foo(true)` is type stable: the value of `condition` is `true`, so that `y = 2.5` and therefore `y` will have type `Float64`. However, values don't participate in multiple dispatch, meaning that Julia's compiler ignores the value of `condition` when inferring `y`'s type. Ultimately, `y` is treated as potentially being either `Int64` or `Float64`, leading to type instability.

The issue in this case can be easily resolved by replacing `1` by `1.0`, thus ensuring that `y` is always `Float64`. More generally, we could employ similar techniques to the first "gotcha", where values are converted to a specific concrete type.

An alternative solution relies on dispatching by value, a technique we already explored and implemented for tuples. This technique makes it possible to communicate information about values directly to the compiler. It's based on the type `Val` in conjunction with the keyword `where` introduced here.

Specifically, for any function `foo` and value `a` that you seek the compiler to know, you need to include `::Val{a}` as an argument. In this way, `a` is interpreted as a type parameter, which can then be identified using the `where` keyword within the function definition. Finally, when calling `foo`, we need pass `Val(a)` as its input.

Applied to our example, type instability in `foo` is caused because the value of `condition` isn't known by the compiler. Dispatching by value provides a mechanism to explicitly convey this information and hence solve the type instability.

```
function foo(condition)
    y = condition ?  2.5  :  1

    return [y * i for i in 1:100]
end

@code_warntype foo(true)          # type UNSTABLE
@code_warntype foo(false)         # type UNSTABLE
```

```
function foo(::Val{condition}) where condition
    y = condition ?  2.5  :  1

    return [y * i for i in 1:100]
end

@code_warntype foo(Val(true))     # type stable
@code_warntype foo(Val(false))    # type stable
```

GOTCHA 5: VARIABLES AS DEFAULT VALUES OF KEYWORD ARGUMENTS

Functions accept both positional and keyword arguments. The possibility of keyword arguments in particular allows the user to assign default values. If these default values are set through variables rather than literal values, a type instability will be introduced. The reason is that such variables will be treated as global variables.

```
foo(; x) = x

β = 1
@code_warntype foo(x=β)          #type stable
```

```
foo(; x = 1) = x


@code_warntype foo()             #type stable
```

```
foo(; x = β) = x

β = 1
@code_warntype foo()             #type UNSTABLE
```

In case you necessarily need to set a variable as a default value, there are still a few strategies you could follow to restore type stability.

One set of solutions leverages the techniques we introduced for global variables. hese include type-annotating the global variable (*Solution 1a*) or defining it as a constant (*Solution 1b*).

Another strategy involves defining a function that stores the default value. By doing so, you can take advantage of type inference, with the function attempting to infer a concrete type for the default value (*Solution 2*).

You can also solve the type instability by adopting a local approach, where type annotations are added to either the keyword argument (*Solution 3a*) or the default value itself (*Solution 3b*). Note that this isn't necessary when positional arguments are used as default values of keyword arguments (*Solution 4*).

All these scenarios are represented below.

```
foo(; x = β) = x

const β = 1
@code_warntype foo()            #type stable
```

```
foo(; x = β) = x

β::Int64 = 1
@code_warntype foo()            #type stable
```

```
foo(; x = β()) = x

β() = 1
@code_warntype foo()            #type stable
```

```
foo(; x::Int64 = β) = x

β = 1
@code_warntype foo()            #type stable
```

```
foo(; x = β::Int64) = x

β = 1
@code_warntype foo()            #type stable
```

```
foo(β; x = β) = x

β = 1
@code_warntype foo(β)           #type stable
```

GOTCHA 6: CLOSURES CAN EASILY INTRODUCE TYPE INSTABILITIES

Closures are a fundamental concept in programming. They refer to functions that capture and retain access to variables from the scope in which they were defined. In practical terms, a closure arises when **one function is defined inside another**, including the case where anonymous functions are used inside a function.

Although closures provide a convenient way to write modular and self-contained code, they can sometimes introduce type instabilities. While Julia has made progress in mitigating these issues, they have persisted for years and remain a source of potential inefficiency. For this reason, it's essential to understand not only the consequences of using closures carelessly, but also to learn strategies for addressing their performance challenges.

CLOSURES ARE COMMON IN CODING

There are several scenarios where closures emerge naturally. One such scenario is when a task requires multiple steps, but you prefer to keep a single self-contained unit of code. For instance, this approach is particularly useful if a function needs to perform multiple interdependent steps, such as data preparation (e.g., setting parameters or initializing variables) and subsequent computations based on that data. By nesting a function within another, you can keep related code organized and contained within the same logical block, promoting code readability and maintainability.

To illustrate how code implements a task with and without closures, we'll use generic code. This isn't intended to be executed, but rather to demonstrate the underlying structure.

```
function task()
        # <here you define parameters and initialize variables>

    function output()
        # <here you do computations with the parameters and variables>
    end

    return output()
end

task()
```

```
function task()
        # <here, you define parameters and initialize variables>

    return output(<variables>, <parameters>)
end

function output(<variables>, <parameters>)
        # <here, you do some computations with the variables and parameters>
end

task()
```

Although the approach using closures may seem more intuitive, it can easily introduce type instability. This occurs when one of these conditions hold:

- variables or arguments are redefined inside the function (e.g., when updating a variable)
- the order in which functions are defined is altered
- anonymous functions are introduced

Each of these cases is explored below, where we refer to the containing function as the *outer function* and the closure as the *inner function*.

WHEN THE ISSUE ARISES

Let's start examining three examples. They cover all the possible situations where closures could result in type instability.

The first examples reveal that the placement of the inner function could matter for type stability.

```
function foo()
    x           = 1
    bar()       = x

    return bar()
end

@code_warntype foo()      # type stable
```

```
function foo()
    bar(x)      = x
    x           = 1

    return bar(x)
end

@code_warntype foo()      # type stable
```

```
function foo()
    bar()       = x
    x           = 1

    return bar()
end

@code_warntype foo()      # type UNSTABLE
```

```
function foo()
    bar()::Int64 = x::Int64
    x::Int64     = 1

    return bar()
end

@code_warntype foo()      # type UNSTABLE
```

```
function foo()
    x = 1

    return bar(x)
end

bar(x) = x

@code_warntype foo()      # type stable
```

The second example establishes that type instability arises when closures are combined with reassignments of variables or arguments. This issue even emerges when the reassignment involves the same object, including trivial expressions such as `x = x`. The example also reveals that type annotating the redefined variable or the closure doesn't resolve the problem.

```
function foo()
    x            = 1
    x            = 1            # or 'x = x', or 'x = 2'

    return x
end

@code_warntype foo()            # type stable
```

```
function foo()
    x            = 1
    x            = 1            # or 'x = x', or 'x = 2'
    bar(x)       = x

    return bar(x)
end

@code_warntype foo()            # type stable
```

```
function foo()
    x            = 1
    x            = 1            # or 'x = x', or 'x = 2'
    bar()        = x

    return bar()
end

@code_warntype foo()            # type UNSTABLE
```

```julia
function foo()
    x::Int64     = 1
    x            = 1
    bar()::Int64 = x::Int64

    return bar()
end

@code_warntype foo()            # type UNSTABLE
```

```julia
function foo()
    x::Int64     = 1
    bar()::Int64 = x::Int64
    x            = 1

    return bar()
end

@code_warntype foo()            # type UNSTABLE
```

```julia
function foo()
    bar()::Int64 = x::Int64
    x::Int64     = 1
    x            = 1

    return bar()
end

@code_warntype foo()            # type UNSTABLE
```

```julia
function foo()
    x            = 1
    x            = 1            # or 'x = x', or 'x = 2'

    return bar(x)
end

bar(x) = x

@code_warntype foo()            # type stable
```

Finally, the last example deals with situations involving multiple closures. It highlights that the order in which you define them could matter for type stability. The third tab in particular demonstrates that passing closures as function arguments can sidestep the issue. However, such an approach is at odds with how code is generally written in Julia.

```julia
function foo(x)
    closure1(x) = x
    closure2(x) = closure1(x)

    return closure2(x)
end

@code_warntype foo(1)            # type stable
```

```julia
function foo(x)
    closure2(x) = closure1(x)
    closure1(x) = x

    return closure2(x)
end

@code_warntype foo(1)            # type UNSTABLE
```

```julia
function foo(x)
    closure2(x, closure1) = closure1(x)
    closure1(x)           = x

    return closure2(x, closure1)
end

@code_warntype foo(1)            # type stable
```

```julia
function foo(x)
    closure2(x) = closure1(x)

    return closure2(x)
end

closure1(x) = x

@code_warntype foo(1)            # type stable
```

In the following, we'll examine specific scenarios where these patterns emerge. The examples reveal that the issue can occur more frequently than we might expect. For each scenario, we'll also provide a solution that enables the use of a closure approach. Nonetheless, if the function captures a performance critical part of your code, it's probably wise to avoid closures.

"BUT NO ONE WRITES CODE LIKE THAT"

i) Transforming Variables through Conditionals

```julia
x = [1,2]; β = 1

function foo(x, β)
    (β < 0) && (β = -β)          # transform 'β' to use its absolute value

    bar(x) = x * β

    return bar(x)
end

@code_warntype foo(x, β)         # type UNSTABLE
```

```julia
x = [1,2]; β = 1

function foo(x, β)
    (β < 0) && (β = -β)          # transform 'β' to use its absolute value

    bar(x,β) = x * β

    return bar(x,β)
end

@code_warntype foo(x, β)         # type stable
```

```julia
x = [1,2]; β = 1

function foo(x, β)
    δ = (β < 0) ? -β : β         # transform 'β' to use its absolute value

    bar(x) = x * δ

    return bar(x)
end

@code_warntype foo(x, β)         # type stable
```

```julia
x = [1,2]; β = 1

function foo(x, β)
    β = abs(β)                     # 'δ = abs(β)' is preferable (you should avoid redefining
variables)

    bar(x) = x * δ

    return bar(x)
end

@code_warntype foo(x, β)         # type stable
```

Recall that the compiler doesn't dispatch by value, and so whether the condition holds is irrelevant. For instance, the type instability would still hold if we wrote `1 < 0` instead of `β < 0`. Moreover, the value used to redefine `β` is also unimportant, with the same conclusion holding if you write `β = β`.

ii) Anonymous Functions inside a Function

Using an anonymous function inside a function is another common form of closure. Considering this, type instability also arises in the example above if we replace the inner function `bar` for an anonymous function. To demonstrate this, we apply `filter` with an anonymous function that keeps all the values in `x` that are greater than `β`.

```
x = [1,2]; β = 1

function foo(x, β)
    (β < 0) && (β = -β)          # transform 'β' to use its absolute value

    filter(x -> x > β, x)        # keep elements greater than 'β'
end

@code_warntype foo(x, β)         # type UNSTABLE
```

```
x = [1,2]; β = 1

function foo(x, β)
    δ = (β < 0) ? -β : β         # define 'δ' as the absolute value of 'β'

    filter(x -> x > δ, x)        # keep elements greater than 'δ'
end

@code_warntype foo(x, β)         # type stable
```

```
x = [1,2]; β = 1

function foo(x, β)
    β = abs(β)                     # 'δ = abs(β)' is preferable (you should avoid redefining
variables)

    filter(x -> x > β, x)        # keep elements greater than β
end

@code_warntype foo(x, β)         # type stable
```

iii) Variable Updates

```julia
function foo(x)
    β = 0                       # or 'β::Int64 = 0'
    for i in 1:10
        β = β + i               # equivalent to 'β += i'
    end

    bar() = x + β               # or 'bar(x) = x + β'

    return bar()
end

@code_warntype foo(1)           # type UNSTABLE
```

```julia
function foo(x)
    β = 0
    for i in 1:10
        β = β + i
    end

    bar(x,β) = x + β

    return bar(x,β)
end

@code_warntype foo(1)           # type stable
```

```julia
x = [1,2]; β = 1

function foo(x, β)
    (1 < 0) && (β = β)

    bar(x) = x * β

    return bar(x)
end

@code_warntype foo(x, β)         # type UNSTABLE
```

iv) The Order in Which you Define Functions Could Matter Inside a Function

To illustrate this claim, suppose you want to define a variable `x` that depends on a parameter `β`. However, `β` is measured in one unit (e.g., meters), while `x` requires `β` to be expressed in a different unit (e.g., centimeters). This implies that, before defining `x`, we must rescale `β` to the appropriate unit.

Depending on how we implement the operation, a type instability could emerge.

```julia
function foo(β)
    x(β)                  =  2 * rescale_parameter(β)
    rescale_parameter(β)  =  β / 10

    return x(β)
end

@code_warntype foo(1)      # type UNSTABLE
```

```julia
function foo(β)
    rescale_parameter(β)  =  β / 10
    x(β)                  =  2 * rescale_parameter(β)

    return x(β)
end

@code_warntype foo(1)      # type stable
```

FOOTNOTES

[1.] A similar problem would occur if we replaced `0` by `0.0` and `x` is an integer.